Methods for merging fragmented facets obtained from point cloud segmentation algorithms

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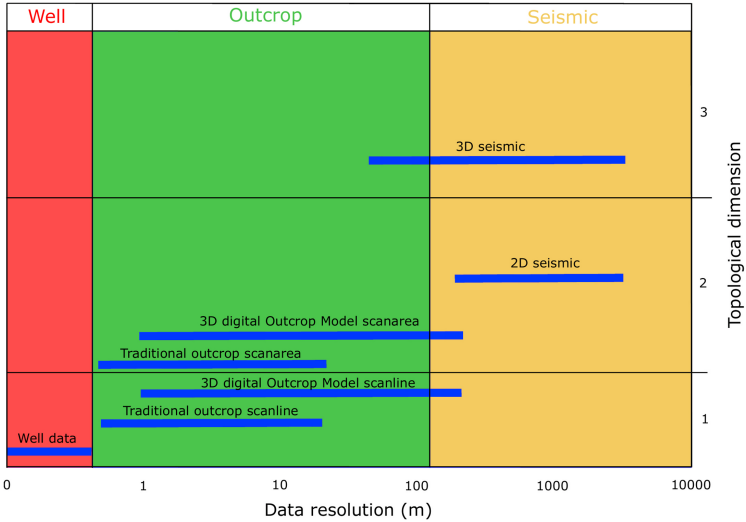
Digital Outcrop Models (DOMs)

DOMs are qualitative three-dimensional representations of suitable outcrops (Bistacchi et al. 2015), filling the resolution gap between well data and seismics.

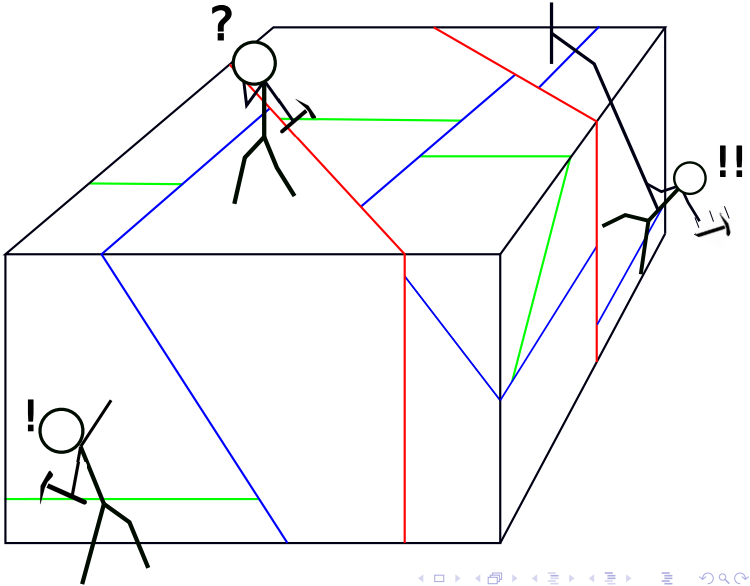


Outcrops are just sections

When analysing an outcrop, we are looking at sections of a volume thus it is impossible to define its 3D properties just by looking at one section

Question

Can we still use outcrops to infer the properties of the volume?



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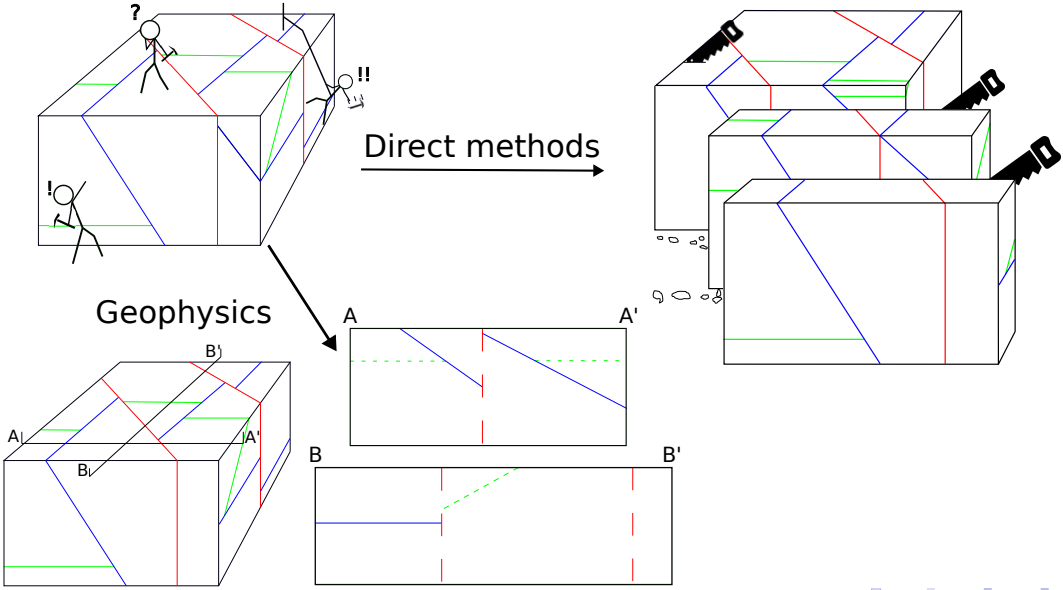


Outcrops are just sections

Possible solution:

1. Study as many optimal "natural sections" (i.e. outcrops)
2. Obtain for each section different characterizing parameters.

Note

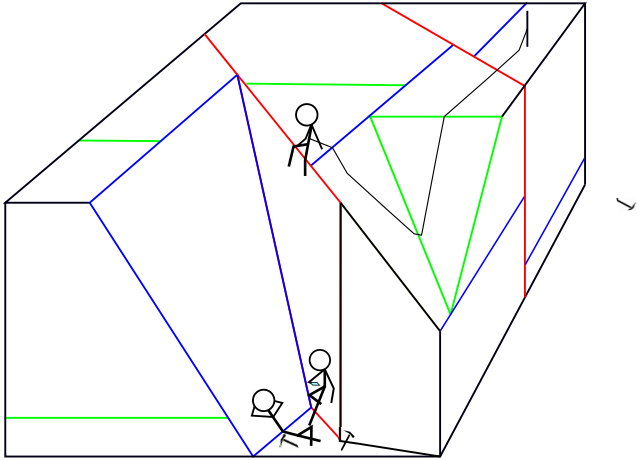
Statistically determine properties of the network using **stochastic** Descrete Fracture Networks (DFNs).

	Parameter	Fracture network	Fracture set
Tectonics	*Number of sets*	✓	
	Deformation mechanism		✓
	Kinematics		✓
	Tectonic phase/stage		✓
Orientation	*Mean dip/direction*		✓
	Dip/direction variance		✓
Topology	*Connectivity*	✓	
	Terminations	✓	
Spacing	*Mean spacing*		✓
	Spacing distribution		✓
Size	*Mean lenght*		✓
	Lenght distribution		✓
	Mean height		✓
	Height distribution		✓
	Height/lenght ratio		✓
Density/intensity	*Mean density*	✓	✓
	Mean intensity	✓	✓
Aperture	*Mean aperture*		✓
	Aperture distribution		✓
	Aperture/lenght ratio		✓
	REV	✓	✓
	Spatial distribution	✓	✓

Fracture facets

Fracture facet

We define as fracture facet the morphological expression of planar discontinuities on an outcrop face.



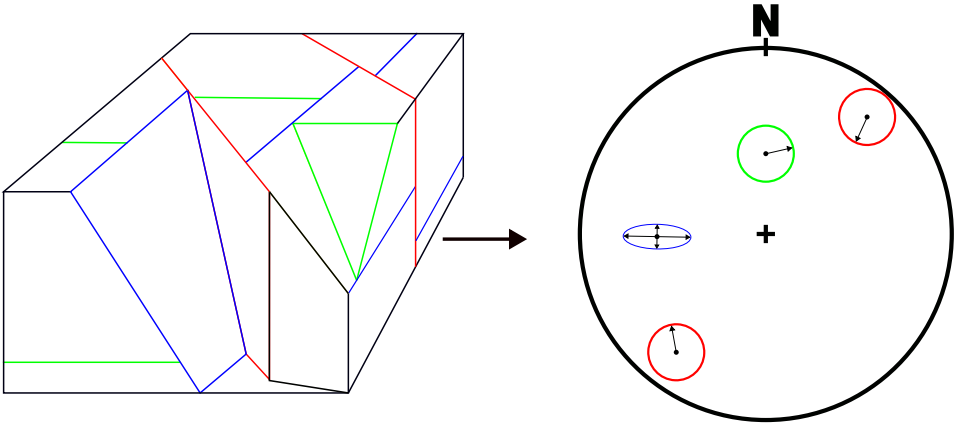
These objects can be used to determine for each fracture set:

1. Orientation distribution
2. Height distribution
3. Vertical topology

The problem: segmentation

Segmentation

Segmentation algorithms divide the point cloud in regions of similar dip and dip direction values. To correctly identify such values **field work needs to be carried out**.



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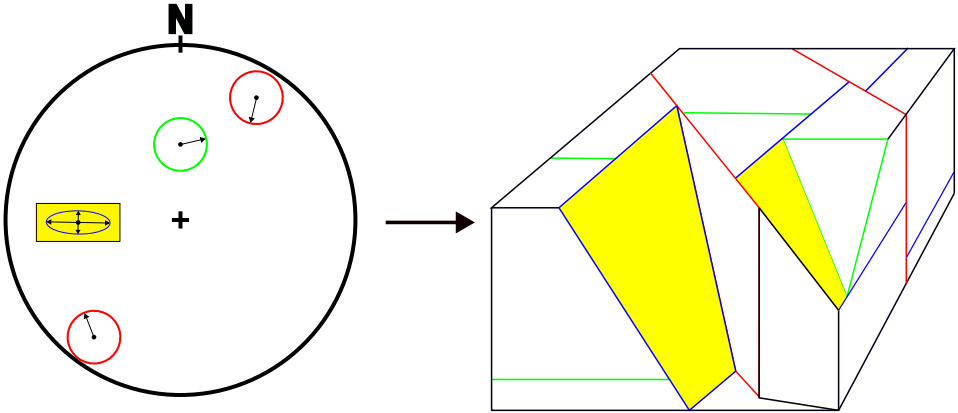
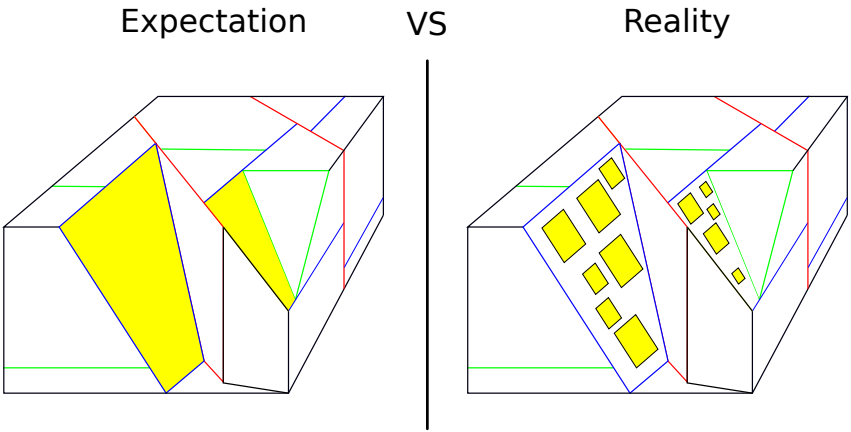


The problem: segmentation

Segmentation algorithms can help in identifying parallel planes but rarely has a clean output.

Question

How can we correct for over-segmentation?



The formula

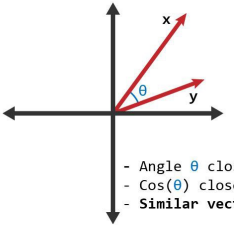
Cosine similarity

Scalar metric that measures the similarity of two vectors (Lee and Jung 2021):

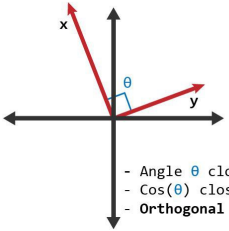
$$\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|a| \cdot |b|}$$

with $\cos\theta \in [-1, 1]$

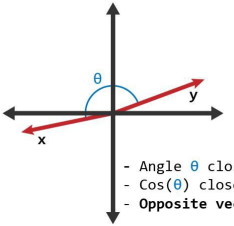
This means that for:



- Angle θ close to 0
- Cos(θ) close to 1
- **Similar vectors**



- Angle θ close to 90
- Cos(θ) close to 0
- **Orthogonal vectors**



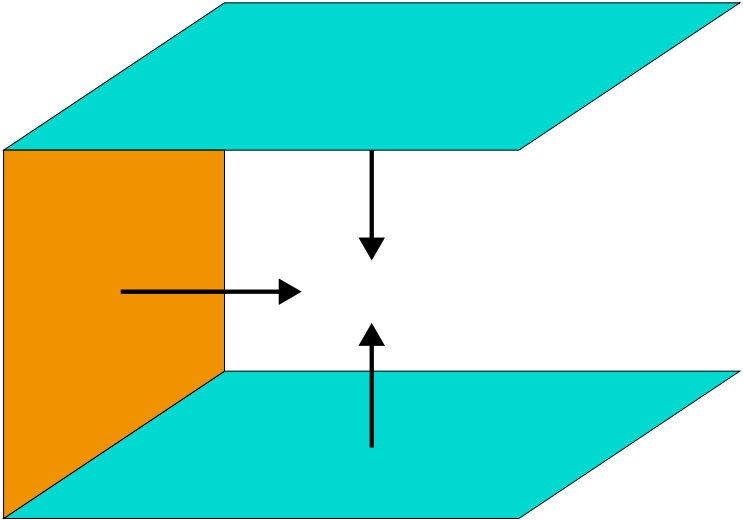
- Angle θ close to 180
- Cos(θ) close to -1
- **Opposite vectors**

Cosine similarity: normals

Cosine similarity between normals define if two facets are dipping in the same direction. For values around 1 and -1 the facets are parallel.

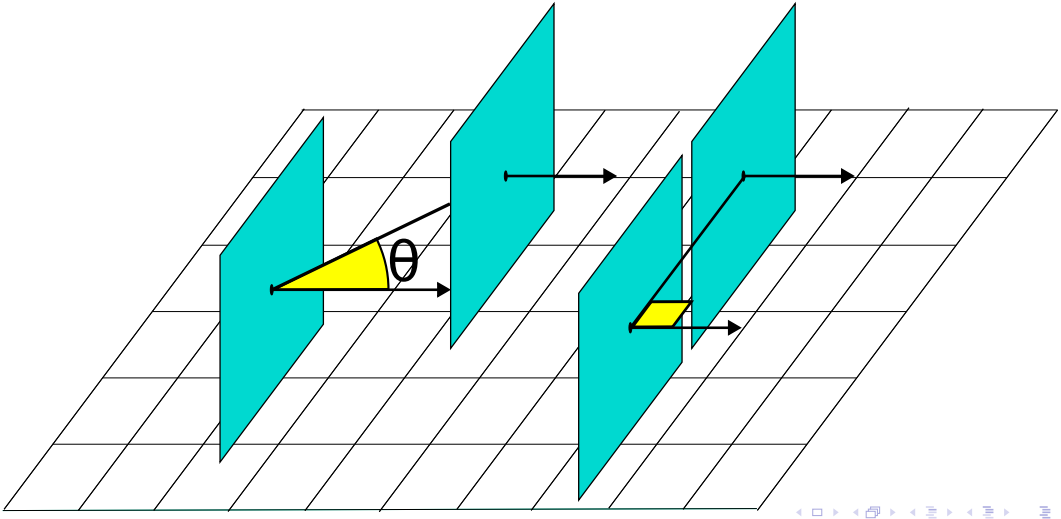
Note

Already taken care of by segmentation; normal vectors represent both the orientation and dip of a face.



Cosine similarity: coplanarity

Cosine similarity between normals of two facets **does not give information about coplanarity** → other vectors need to be used.

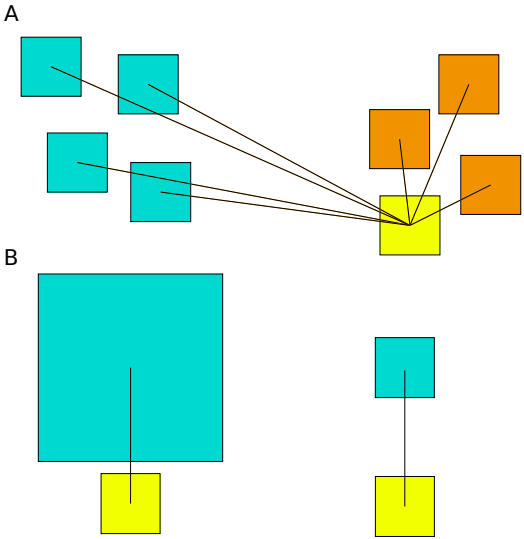


Euclidean distance

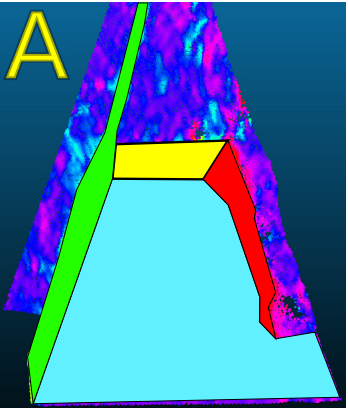
Coplanarity is not enough. Two facets can be parallel and coplanar but very distant from each other.

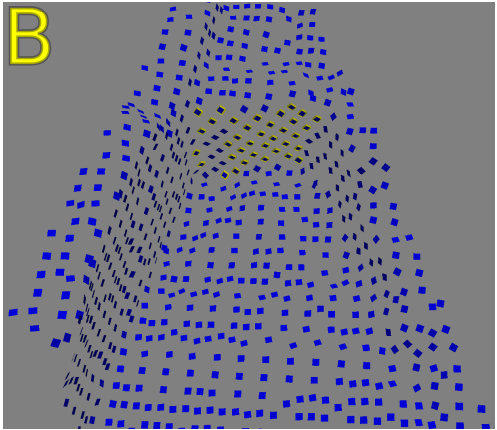
Note

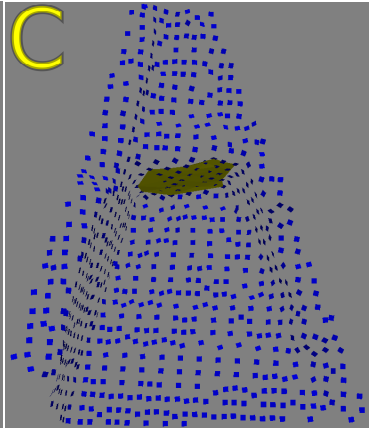
Metric valid only when facets are regular and of the same size; comparing two differently sized facets will result in an erroneous estimation of the distance.



Results







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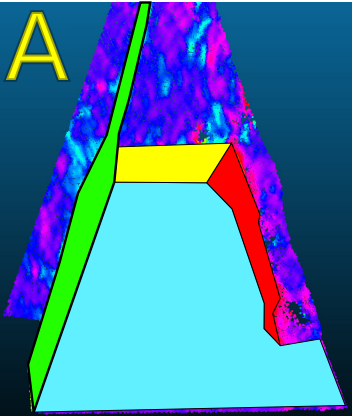

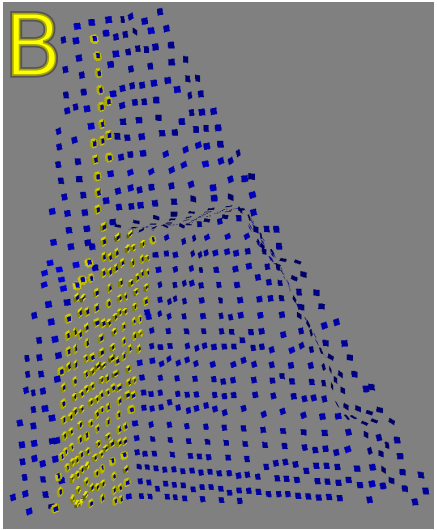

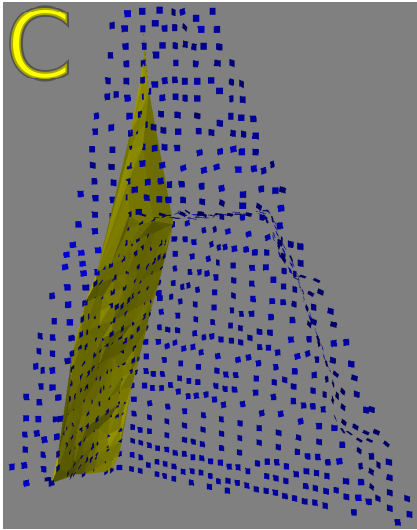

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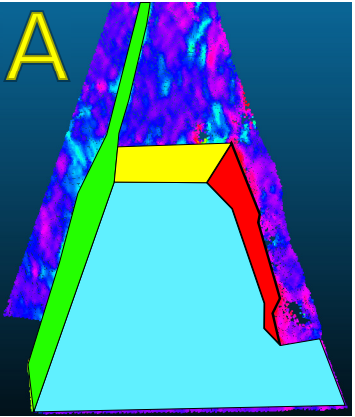

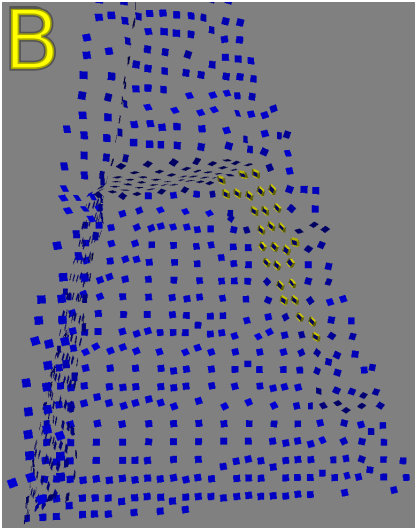

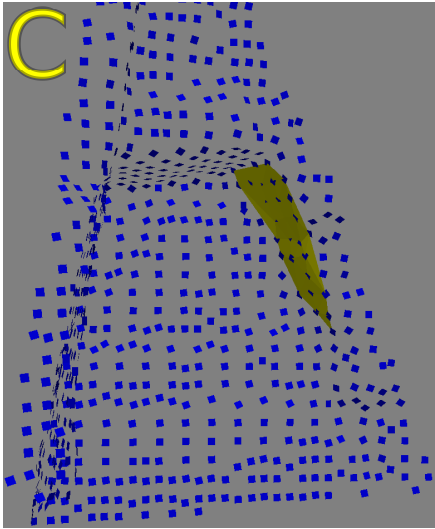

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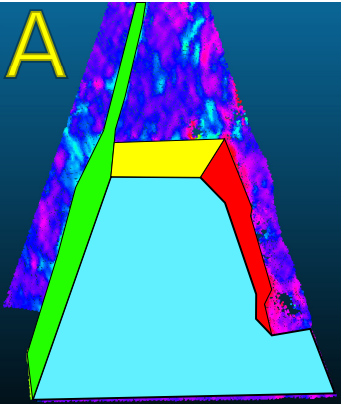

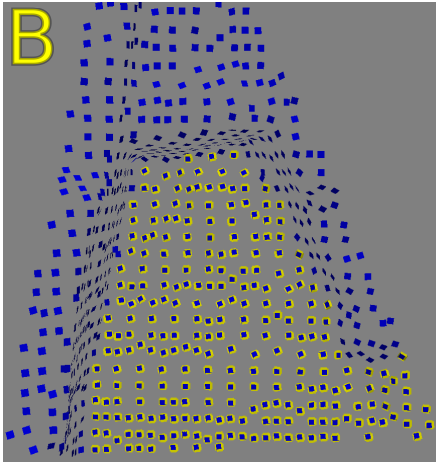

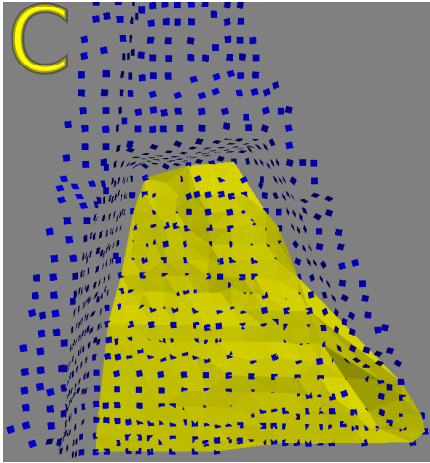

Conclusions and references

- Cosine similarity as new scalar to measure:
  1. Orientation
  2. Coplanarity
- Field observation and data gathering are necessary to define fracture sets and postulate geological models to guide the digital interpretation $\rightarrow$ fieldwork is fundamental.
- For the future:
  1. Automatic parameters calibration.
  2. Automatic guided merging methods.

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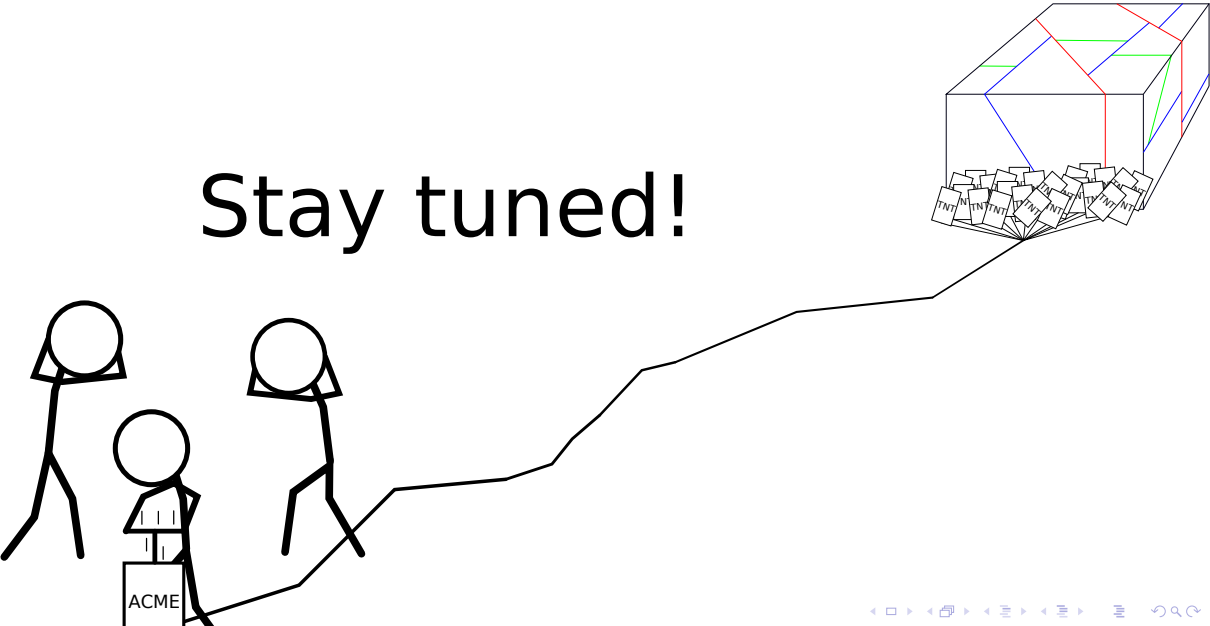
[1] A Bistacchi et al. "Photogrammetric Digital Outcrop Reconstruction, Visualization with Textured Surfaces, and Three-Dimensional Structural Analysis and Modeling: Innovative Methodologies Applied to Fault-Related Dolomitization (Vajont Limestone, Southern Alps, Italy)". In: *Geosphere* 11.6 (Dec. 2015), pp. 2031–2048. ISSN: 1553-040X, 1553-040X. DOI: 10.1130/GES01005.1. (Visited on 09/03/2022).

[2] Hongjae Lee and Jiyoung Jung. "Clustering-Based Plane Segmentation Neural Network for Urban Scene Modeling". In: *Sensors* 21.24 (Dec. 2021), p. 8382. ISSN: 1424-8220. DOI: 10.3390/s21248382. (Visited on 08/20/2023).

Stay tuned!